

CSE421 – Computer Networks

Assignment 3: Computer Network Ethics

Based on Lecture 0.3: Introduction to Computer Network Ethics
Department of Computer Science and Engineering, BRAC University

Instructions

Each question presents a realistic scenario related to computer network ethics. For each question:

- Write a short essay (100-400 words) demonstrating your reasoning process.
- Provide concrete examples or original analogies whenever possible.
- Where possible, relate questions to your course learning or personal/professional experiences.
- Each question carries a weightage of 10 marks, resulting in a total of 50 marks for the entire assignment.
- Write your assignment in word processing software (MS word or OpenOffice or Latex) and submit it in .pdf format.
- Avoid copying from textbooks, slides, or online sources (and classmates!). Your answers should clearly reflect your own thinking style and analytical approach. If plagiarism is found, you will receive a zero.
- Your assignment must be free of AI-generated content. Do not use ChatGPT or any other large language model to produce your answers. Therefore, your assignment file must have an AI-generated text score of 20% or lower on Turnitin.
- Assignment Submission link: <https://forms.gle/s6iQgyU1R2ehMnae6>
- Deadline: 11:59 pm, 25 April, 2026

Question 1

You are a network administrator at a mid-sized Bangladeshi company. A colleague accidentally discovers confidential client data (including national ID numbers, phone numbers, and salary details) due to a misconfigured firewall rule. Your colleague tells you informally over lunch instead of filing a report. What steps should you take—both ethically and professionally—and why? In your answer, discuss the relevant ethical principles (privacy, responsibility, integrity) and the potential consequences of inaction.

Question 2

A group of students at a Bangladeshi university creates a Wi-Fi hotspot using the university's network and shares the password on social media, allowing outsiders (non-students) to connect freely. During exam week, the network becomes extremely slow, and the online exam portal crashes for 15 minutes. Analyse the ethical issues involved. Who is responsible, and what principles have been violated?

Question 3

A student notices that the university's Learning Management System (LMS) has a bug: by modifying the URL parameters, they can view other students' submitted assignments, grades, and personal information. The student does not exploit this for personal gain but shares the discovery on a public Facebook group, exposing the vulnerability to thousands of people before the university can fix it. Evaluate the student's actions from an ethical standpoint. Was the disclosure responsible?

Question 4

Two friends in a university hostel share a laptop. One friend, without the other's knowledge, installs a keylogger to capture the other's passwords—including their email, LMS, and mobile banking credentials. The friend claims he "just wanted to see if it worked" as part of learning about cybersecurity. Analyse the ethical and legal dimensions of this action. Does the educational intent justify the means?

Question 5

A junior developer at a software company in Bangladesh is asked to build a user registration system. The developer stores all user passwords in plain text in the database because "it's faster to develop" and "the project deadline is tomorrow." A month later, the database is breached, and thousands of users' passwords are leaked. Evaluate the developer's decision from the perspective of professional ethics, competence, and responsibility.